

Docker Assignment: 3

Docker Networking:

```
python-backend > Dockerfile
1 FROM python:3.11-slim
2 WORKDIR /app
3 COPY . .
4 RUN pip install -r requirements.txt

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -d --name container2 alpine sleep infinity sh
990d6c8737e9be1734c34bee754f4cf179a5d98dcd1e9b4f3cb1ca0c2fe66a5f
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
990d6c8737e9 alpine "sleep infinity sh" 7 seconds ago Up 6 seconds container2
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -d --name container1 alpine sleep infinity sh
d2fd84c9097a60e48d8ad4029968b462d1790e65d0c98aad988349c46429f306
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
d2fd84c9097a alpine "sleep infinity sh" 7 seconds ago Up 7 seconds container1
990d6c8737e9 alpine "sleep infinity sh" 35 seconds ago Up 34 seconds container2
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker network inspect container1
[]
Error response from daemon: network container1 not found
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker inspect container1
[
  {
    "Id": "d2fd84c9097a60e48d8ad4029968b462d1790e65d0c98aad988349c46429f306",
    "Created": "2026-07-27T18:13:59.061341052Z",
    "Path": "sleep",
    "Args": [
      "infinity",
      "sh"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 972,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-07-27T18:13:31.881058249Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b",
    "ResolveConfPath": "/var/lib/docker/containers/990d6c8737e9be1734c34bee754f4cf179a5d98dcd1e9b4f3cb1ca0c2fe66a5f/resolve.conf"
  }
]

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker inspect container2
[
  {
    "Id": "990d6c8737e9be1734c34bee754f4cf179a5d98dcd1e9b4f3cb1ca0c2fe66a5f",
    "Created": "2026-07-27T18:13:31.792852936Z",
    "Path": "sleep",
    "Args": [
      "infinity",
      "sh"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 972,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-07-27T18:13:31.881058249Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f8b",
    "ResolveConfPath": "/var/lib/docker/containers/990d6c8737e9be1734c34bee754f4cf179a5d98dcd1e9b4f3cb1ca0c2fe66a5f/resolve.conf"
  }
]
```



```

"WorkingDir": "/",
"Entrypoint": null,
"Labels": {}
},
"NetworkSettings": {
  " SandboxID": "93d2265de082534e58f3baa3b95b261154538a1424b714f70b8653ba527c7525",
  "SandboxKey": "/var/run/docker/netns/93d2265de082",
  "Ports": {},
  "Networks": {
    "appnetwork": {
      "IPAMConfig": null,
      "Links": null,
      "Aliases": null,
      "DriverOpts": null,
      "GwPriority": 0,
      "NetworkID": "c5f8540d4a331cf0e095b680f0f355987e2d799dc19756571dc7f5ee3d2b9891",
      "EndpointID": "d807549c7303cc0e6b2581d40639f734cf782608de67fbca4d2c87324e25b0d",
      "Gateway": "172.18.0.1",
      "IPAddress": "172.18.0.2",
      "MacAddress": "3a:39:d4:0a:d9:ed",
      "IPPrefixLen": 16,
      "IPv6Gateway": "",
      "GlobalIPv6Address": "",
      "GlobalIPv6PrefixLen": 0,
      "DNSNames": [
        "container1",
        "438f588e09b0"
      ]
    }
  }
}

```

```
1 FROM python:3.11-slim
```

```
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker inspect container2
```

```
{
  "Id": "e46ab404b271bb78171ef53b0f0cb2ff000256d5921b3b1f356553347c9ef0be",
  "Created": "2026-07-28T15:07:39.556729731Z",
  "Path": "sh",
  "Args": [],
  "State": {
    "Status": "running",
    "Running": true,
    "Paused": false,
    "Restarting": false,
    "OOMKilled": false,
    "Dead": false,
    "Pid": 476,
    "ExitCode": 0,
    "Error": "",
    "StartedAt": "2026-07-28T15:07:39.639711188Z",
    "FinishedAt": "0001-01-01T00:00:00Z"
  },
  "Image": "sha256:28bd5fe8b56d1bd048e5babf5b10710ebe0bae67db86916198a6eec434943f0b",
  "ResolvConfPath": "/var/lib/docker/containers/e46ab404b271bb78171ef53b0f0cb2ff000256d5921b3b1f356553347c9ef0be/resolv.conf",
  "HostnamePath": "/var/lib/docker/containers/e46ab404b271bb78171ef53b0f0cb2ff000256d5921b3b1f356553347c9ef0be/hostname",
  "HostsPath": "/var/lib/docker/containers/e46ab404b271bb78171ef53b0f0cb2ff000256d5921b3b1f356553347c9ef0be/hosts",
  "LogPath": "/var/lib/docker/containers/e46ab404b271bb78171ef53b0f0cb2ff000256d5921b3b1f356553347c9ef0be/e46ab404b271bb78171ef53b0f0cb2ff000256d5921b3b1f356553347c9ef0be-json.log",
  "Name": "/container2",
  "RestartCount": 0,
  "Driver": "overlays",

```

```

]
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker exec -it container1 sh
/ # ping 172.18.0.3
PING 172.18.0.3 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.515 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.121 ms
64 bytes from 172.18.0.3: seq=2 ttl=64 time=0.230 ms
64 bytes from 172.18.0.3: seq=3 ttl=64 time=0.129 ms
^C
--- 172.18.0.3 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.121/0.248/0.515 ms
/ # ping container2
PING container2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.111 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.129 ms
64 bytes from 172.18.0.3: seq=2 ttl=64 time=0.074 ms
64 bytes from 172.18.0.3: seq=3 ttl=64 time=0.128 ms
64 bytes from 172.18.0.3: seq=4 ttl=64 time=0.124 ms
64 bytes from 172.18.0.3: seq=5 ttl=64 time=0.203 ms
64 bytes from 172.18.0.3: seq=6 ttl=64 time=0.118 ms
64 bytes from 172.18.0.3: seq=7 ttl=64 time=0.124 ms
64 bytes from 172.18.0.3: seq=8 ttl=64 time=0.143 ms
^C
--- container2 ping statistics ---
9 packets transmitted, 9 packets received, 0% packet loss
round-trip min/avg/max = 0.074/0.128/0.203 ms
/ # exit

```

```
1 FROM python:3.11-slim

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker network create othernetwork
17e5d14836f27db649efd49ff7ba06d373911d15624346c32c1850b83527ebd6
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
c5f8540d4a33        appnetwork          bridge              local
ce879328186c        bridge              bridge              local
95ff2c9e9c9d        host                host                local
9af25ca88bb0        none                null                local
17e5d14836f2        othernetwork        bridge              local
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker run -dit --name container3 --network othernetwork alpine sh
72b0a96ae38647ab35b22dff4deef9645fe07679741ebe8feaa08f037cb3a6
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker ps -a
CONTAINER ID        IMAGE               COMMAND              CREATED            STATUS              PORTS              NAMES
72b0a96ae386        alpine             "sh"                 13 seconds ago    Up 12 seconds      0.0.0.0:80->8000/tcp container3
e46ab404b271        alpine             "sh"                 10 minutes ago    Up 10 minutes      container2
438f588e09b0        alpine             "sh"                 10 minutes ago    Up 10 minutes      container1
3bb60bde562e        python-backend:v1  "uvicorn main:app --" 25 hours ago      Exited (255) 21 hours ago
8e6327237169        python-backend:v1  "uvicorn main:app --" 25 hours ago      Created
3a4105dda2f9        python-backend:v1  "uvicorn main:app --" 26 hours ago      Exited (0) 25 hours ago
9218b79bb441        python-backend:v1  "uvicorn main:app --" 26 hours ago      Created
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker inspect container3
[
  {
    "Id": "72b0a96ae38647ab35b22dff4deef9645fe07679741ebe8feaa08f037cb3a6",
    "Created": "2026-07-28T15:17:46.959864192Z",
    "Path": "sh",
    "Args": [],
    "State": {
      "Status": "running",
      "Labels": {}
    },
    "NetworkSettings": {
      "SandboxID": "26bd8ec069b99738fc4d950d952e92e1f3b9fd4249d72870b1161c6f1cd4efc2",
      "SandboxKey": "/var/run/docker/netns/26bd8ec069b9",
      "Ports": {},
      "Networks": {
        "othernetwork": {
          "IPAMConfig": null,
          "Links": null,
          "Aliases": null,
          "DriverOpts": null,
          "GwPriority": 0,
          "NetworkID": "17e5d14836f27db649efd49ff7ba06d373911d15624346c32c1850b83527ebd6",
          "EndpointID": "ebe465708f1471e147ccb1da345175f430ac516eae6253d3d6a506bb1ebf67a1",
          "Gateway": "172.19.0.1",
          "IPAddress": "172.19.0.2",
          "MacAddress": "b6:a4:64:8f:ef:b6",
          "IPPrefixLen": 16,
          "IPv6Gateway": "",
          "GlobalIPv6Address": "",
          "GlobalIPv6PrefixLen": 0,
          "DNSNames": [
            "container3",
            "72b0a96ae386"
          ]
        }
      }
    }
  }
]

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker exec -it container1 sh
/ # ping container3
ping: bad address 'container3'
/ # ping 172.19.0.2
PING 172.19.0.2 (172.19.0.2): 56 data bytes
exit
^C
--- 172.19.0.2 ping statistics ---
357 packets transmitted, 0 packets received, 100% packet loss
/ # exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image + docker debug container1
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker network connect othernetwork container1
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker inspect container1
[
  {
    "Id": "438f588e09b0d967c71fc8d5a1be912b83f06ae6b17343506258d32c1bf675c",
    "Created": "2026-07-28T15:07:30.380510993Z",
    "Path": "sh",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Labels": {}
    },
    "NetworkSettings": {
      "SandboxID": "26bd8ec069b99738fc4d950d952e92e1f3b9fd4249d72870b1161c6f1cd4efc2",
      "SandboxKey": "/var/run/docker/netns/26bd8ec069b9",
      "Ports": {},
      "Networks": {
        "othernetwork": {
          "IPAMConfig": null,
          "Links": null,
          "Aliases": null,
          "DriverOpts": null,
          "GwPriority": 0,
          "NetworkID": "17e5d14836f27db649efd49ff7ba06d373911d15624346c32c1850b83527ebd6",
          "EndpointID": "ebe465708f1471e147ccb1da345175f430ac516eae6253d3d6a506bb1ebf67a1",
          "Gateway": "172.19.0.1",
          "IPAddress": "172.19.0.2",
          "MacAddress": "b6:a4:64:8f:ef:b6",
          "IPPrefixLen": 16,
          "IPv6Gateway": "",
          "GlobalIPv6Address": "",
          "GlobalIPv6PrefixLen": 0,
          "DNSNames": [
            "container1",
            "438f588e09b0d967c71fc8d5a1be912b83f06ae6b17343506258d32c1bf675c"
          ]
        }
      }
    }
  }
]
```

```
1 FROM python:3.11-slim

},
"NetworkSettings": {
  "SandboxID": "93d2265de082534e58f3baa3b95b261154538a1424b714f70b8653ba527c7525",
  "SandboxKey": "/var/run/docker/netns/93d2265de082",
  "Ports": {},
  "Networks": {
    "appnetwork": {
      "IPAMConfig": null,
      "Links": null,
      "Aliases": null,
      "DriverOpts": null,
      "GwPriority": 0,
      "NetworkID": "c5f8540d4a331cf0e05b680f0f355987e2d799dcb19756571dc7f5ee3d2b9891",
      "EndpointID": "d887549c7383cc0e6b2581d40639f734cf782608de67bfca4d2c87324e25b0d",
      "Gateway": "172.18.0.1",
      "IPAddress": "172.18.0.2",
      "MacAddress": "3a:39:d4:0a:d9:ed",
      "IPPrefixLen": 16,
      "IPv6Gateway": "",
      "GlobalIPv6Address": "",
      "GlobalIPv6PrefixLen": 0,
      "DNSNames": [
        "container1",
        "438f588e09b0"
      ]
    },
    "othernetwork": {
      "IPAMConfig": {},
      "Links": null,
    }
  }
}

}

PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend> docker exec -it container1 sh
/ # ping 172.19.0.3
PING 172.19.0.3 (172.19.0.3): 56 data bytes
64 bytes from 172.19.0.3: seq=0 ttl=64 time=0.271 ms
64 bytes from 172.19.0.3: seq=1 ttl=64 time=0.148 ms
64 bytes from 172.19.0.3: seq=2 ttl=64 time=0.170 ms
64 bytes from 172.19.0.3: seq=3 ttl=64 time=0.146 ms
^C
--- 172.19.0.3 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.146/0.183/0.271 ms
/ # ping container3
PING container3 (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=0.517 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.231 ms
64 bytes from 172.19.0.2: seq=2 ttl=64 time=0.228 ms
64 bytes from 172.19.0.2: seq=3 ttl=64 time=0.260 ms
^C
--- container3 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.228/0.309/0.517 ms
/ # exit

What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug container1
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\Acer\SRE-DEVOPS\DOCKER\python-backend\python-backend>
```